Interpolation Problem

Polymath Jr

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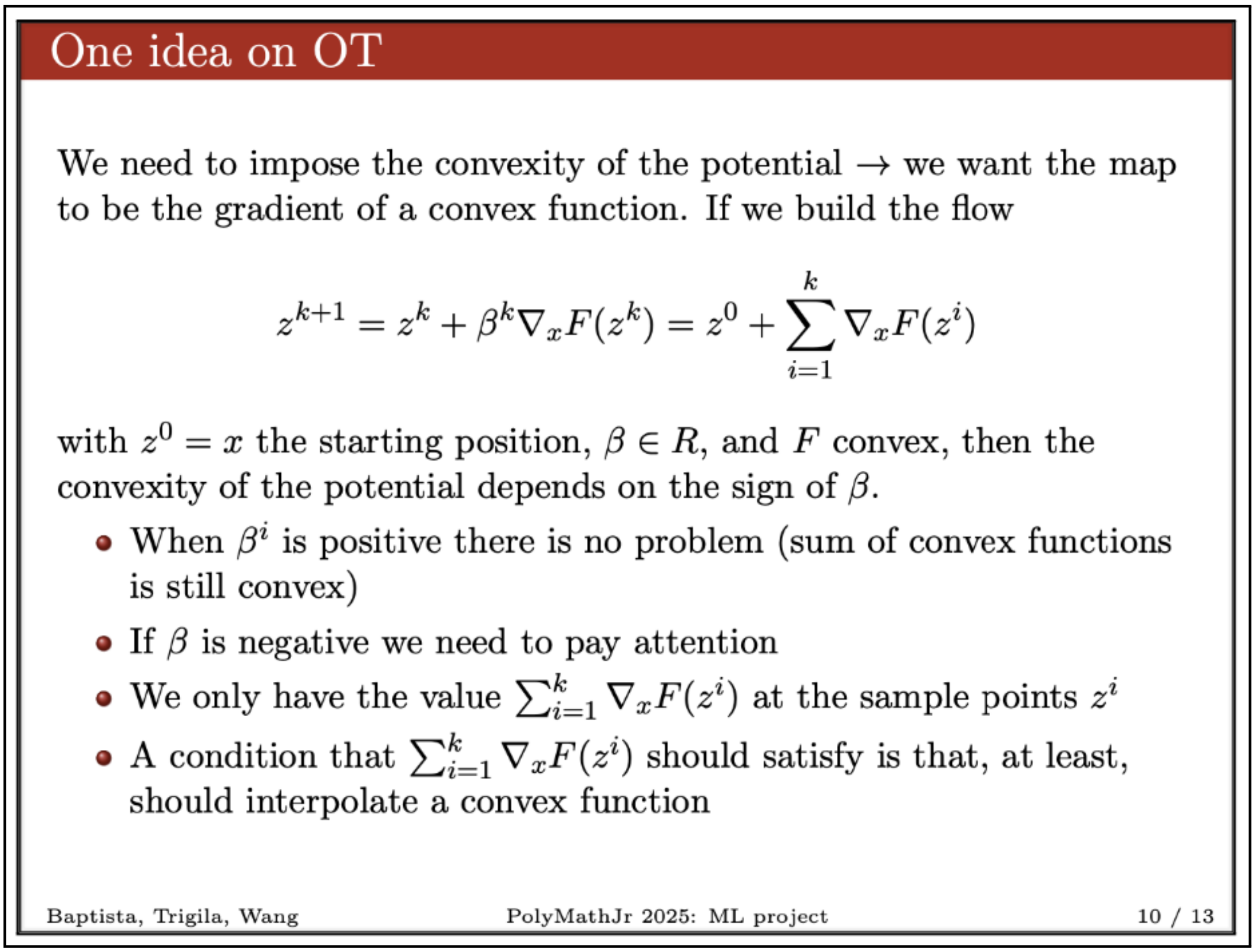

In the slide, when β^i is positive, the sum $\sum \nabla_x F(z^i)$ remains convex — no issue arises. But if any $\beta^i < 0$, convexity is no longer guaranteed, so we need to check whether the resulting sum still corresponds to a convex potential. That's exactly where Xuan's proposed solution to the interpolation problem comes in: it provides a way to verify whether a given sum of gradients (even with negative coefficients) still defines a convex function.

Xuan's Proposed Solution to the Interpolation Problem:

Questions: Given a bunch of points $(x_i.y_i)_{i=1,\ldots,n}$, with $x_i \in \mathbb{R}^d, y_i \in \mathbb{R}$, can this interpolate a convex function?

Idea: Points $(x_i.y_i)$ can interpolate a convex function means that there exist a convex function $f : \mathbb{R}^d \to \mathbb{R}$ such that $f(x_i) = y_i$. We have such convex function if and only if for all $i = 1, ..., n$, there exist subgradient vector $g_i \in \mathbb{R}^d$ such that for all $j = 1, ..., n$

$$y_j \geqslant y_i + g_i^T (x_j - x_i)$$

Defining the Subgradient

If f is **differentiable** at x_0, then the gradient $\nabla f(x_0)$ is the **unique** subgradient.

But if f is **not differentiable**, then there may be **many** valid subgradients — forming a **subdifferential set**, denoted:

$$\partial f(x_0) = \left\{ g \in \mathbb{R}^d \,\middle|\, f(x) \geq f(x_0) + g^\top (x - x_0) \text{ for all } x \right\}$$

Example

Let $f(x, y) = x^2 + 2y$, which is a convex function.

Let $x_i = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, so

$$f(x_i) = 1^2 + 2 \cdot 2 = 5$$

Since f is differentiable, the subgradient at x_i is the gradient:

$$\nabla f(x, y) = \begin{bmatrix} 2x \\ 2 \end{bmatrix} \Rightarrow \nabla f(1, 2) = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

So $g_i = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$.

Now let $x_j = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$, so

$$f(x_j) = 2^2 + 2 \cdot 4 = 4 + 8 = 12$$

Compute the subgradient inequality:

$$x_j - x_i = \begin{bmatrix} 2 \\ 4 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$g_i^\top (x_j - x_i) = \begin{bmatrix} 2 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 2 \cdot 1 + 2 \cdot 2 = 6$$

$$f(x_i) + g_i^\top (x_j - x_i) = 5 + 6 = 11$$

Compare:

$$f(x_j) = 12 \geq 11 = f(x_i) + g_i^\top (x_j - x_i)$$

Therefore, the inequality holds and g_i is a valid subgradient.

This means for every point (x_i, y_i), there is a supporting hyperplane that passes through (x_i, y_i) and keeps all other points on or above it. (Prove: the necesary condition already follows from definition of that convex function. For sufficient condition: we can build a convex function $f(x) = \max_{i=1,..,n}\{y_i + g_i^T (x - x_i)\}$, and $f(x_i) = y_i$ since we have for all $k \neq i$, $y_k + g_k^T (x_i - x_k) \leqslant y_i$.)

Understanding the Convex Interpolation Construction

Given subgradients $g_i \in \mathbb{R}^d$ and values $y_i \in \mathbb{R}$ at points $x_i \in \mathbb{R}^d$, the goal is to construct a convex function that interpolates these values.

A candidate convex function is given by the expression:

$$f(x) = \max_{i=1,\ldots,n} \left\{ y_i + g_i^\top (x - x_i) \right\}.$$

This formula defines a function as the pointwise maximum of a finite collection of affine functions. Now, we will explain why this construction is valid.

Interpretation of Each Term

Each inner term $y_i + g_i^\top (x - x_i)$ defines a hyperplane that passes through the point (x_i, y_i) with slope g_i. For every data point (x_i, y_i), the corresponding affine function represents a local linear lower approximation of the convex function.

Role of the Maximum

Taking the maximum across these affine functions ensures that the resulting function lies above all the supporting hyperplanes. Geometrically, this creates a surface resembling a "tent roof" formed by the upper envelope of the individual planes. The surface bends upward wherever one affine piece overtakes another, preserving convexity.

Why the Resulting Function is Convex

A standard result in convex analysis states that the pointwise maximum of convex functions is itself convex. Since each affine function $y_i + g_i^\top (x - x_i)$ is convex, the function $f(x)$ constructed in this pointwise way is convex by use of this standard result.

Conclusion

This construction demonstrates the sufficiency of the subgradient condition: if the inequality

$$y_j \geq y_i + g_i^\top (x_j - x_i) \quad \text{for all } i, j$$

is satisfied, then one can explicitly construct a convex function $f(x)$ that interpolates all the points (x_i, y_i) by taking the pointwise maximum of the associated affine functions.

Now using this idea, we can construct a linear program: For $i, j = 1, \ldots, n$,

$$y_j \geqslant y_i + g_i^T (x_j - x_i)$$

where $x_i, x_j \in \mathbb{R}^d, y_i, y_j \in \mathbb{R}$ are known vectors and values. And only g_i's are unknown vectors. If this linear program is feasible, then we know there exist such subgradient vectors, so points can interpolate a convex function. If the linear program is not feasible, then points can not interpolate convex functions.